

Driving Next to Automated Vehicles: Emergent Human-machine Cooperation in Mixed Traffic

Yutong Zhang
yuz192@pitt.edu
University of Pittsburgh
Pittsburgh, PA, United States

Shihong Ling
shl282@pitt.edu
University of Pittsburgh
Pittsburgh, PA, United States

Edmond Awad
e.awad@exeter.ac.uk
University of Exeter
Exeter, United Kingdom

Morgan R. Frank
mrfrank@pitt.edu
University of Pittsburgh
Pittsburgh, PA, United States

Na Du
na.du@pitt.edu
University of Pittsburgh
Pittsburgh, PA, United States

ABSTRACT

Introducing automated vehicles (AVs) on public roads may challenge established norms as drivers in human-driven vehicles (HVs) learn to interact with AVs. Our study utilizes a game theory framework to investigate how driver and vehicle driving styles (aggressive vs. conservative), interaction types (HV-HV vs. HV-AV), scenario types (chicken game vs. public goods game scenarios), and time constraints (high vs. low) influence human drivers' decision-making in mixed-traffic environments. According to an online survey study, drivers with aggressive driving styles and high time constraints were more likely to take aggressive actions. More importantly, there were significant interaction effects between vehicle driving styles and scenario types, between scenario types and time constraints, and between driver driving styles and interaction types on driver decision-making. Our findings provide essential insights into the design of AVs and promote the development of related laws and policies to facilitate human-machine cooperation in mixed-traffic environments.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

automated vehicles; mixed-traffic environment; human-machine cooperation; game theory

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1 INTRODUCTION

With the maturity of automated driving technology and the proliferation of automated vehicles (AVs), the future of transportation will involve a combination of AVs, conventional human-driven vehicles (HVs), and other road users (e.g., pedestrians and cyclists). In mixed-traffic environments, HVs have to frequently interact with AVs, including but not limited to one-on-one and AV platoon. Human drivers mostly navigate driving scenarios without incidents through social norms and signaling [31]. However, introducing AVs on public roads may challenge established norms as HVs learn to interact with AVs. How will humans behave as HVs interact with AVs in mixed-traffic scenarios? Will HVs become more cooperative through trust in rule-following AVs? Or will HVs take advantage of AVs for their personal benefit thus decreasing cooperation even with other AVs?

Existing studies have explored the interaction between drivers in HVs and AVs in mixed-traffic environments. Research showed that human behaviors and decisions in HVs were influenced by different factors, such as interaction types (HV-HV vs. HV-AV) [8, 16, 17, 19, 20, 29, 30, 44] and driver driving styles [19]. However, there are some research gaps that need to be addressed. First, AVs have been assumed to be conservative without the possibility for AVs to adapt their behavior based on interactions with other vehicles. Existing research lacks insights into how the variable, AV driving style, impacts human decision-making, and vice versa. Second, although studies have included multiple driving scenarios, none of them theoretically categorize driving scenarios and investigate their effects on drivers' decision-making in mixed-traffic environments. Insights here may connect directly to a wealth of game theoretic research with the potential to influence policy making while optimizing transportation safety and efficiency. Third, existing literature focuses mostly on one or two variables at a time. However, variables may interact to reveal higher-order outcomes in driving scenarios.

In this study, we filled these gaps and investigated how the driver and vehicle driving styles, interaction types, scenario types, and time constraints influenced drivers' decision-making in mixed-traffic environments. We identified two types of interaction in mixed-traffic environments: HV-AV and HV-HV, and assumed that all the interacting vehicles had two distinctly different driving styles: aggressive and conservative. Driving scenarios were developed based on game theoretic interactions and task payoffs were set in

the form of time constraints. We examined drivers' decision-making in different conditions using an online survey study. Our sample consisted of 252 participants recruited from the CloudResearch platform. Our findings will assist relevant practitioners to improve the design of AVs and formulate laws and policies for safe, reliable, and effective HV-AV interaction in mixed-traffic environments.

2 RELATED WORK

Human behaviors and decision-making in mixed-traffic environments have been widely studied, including the interaction between pedestrians and AVs and between drivers in HVs and AVs. Research on pedestrian-AV interaction showed that pedestrians' crossing behaviors were influenced by AV driving behaviors [13, 28, 48], communication methods [1, 2, 37, 45], road conditions [13], and individuals' characteristics [1, 6, 26]. For example, Colley et al. [2] found that AV external communication, other pedestrians' behavior, and previous experience influenced pedestrians' crossing decisions. Zhao et al. [48] found that pedestrians showed a greater intention to engage in risky road-crossing behaviors in front of AVs compared to HVs. For HV-AV interaction, prior work has shown that interaction types (HV-HV vs. HV-AV) [8, 16, 17, 19, 20, 29, 30, 44], communication methods [3, 34], driver driving styles [19], time constraints [44], and expectations for AVs [22] influenced human drivers' behaviors and decision-making.

Some studies investigated the effect of interaction types on the decision-making of human drivers through web-based surveys [17, 19], driving simulator studies [8, 16, 29, 38, 44], and field experiments [20, 30]. For example, a cross-national survey study [17] indicated that human drivers were more inclined to bully AVs than HVs. Ma and Zhang [19] explored how interaction types and driver driving styles influenced human decision-making and subjective feelings. The results suggested that aggressive and moderate drivers were more likely to exhibit aggressive behavior in HV-AV interaction than in HV-HV interaction, and aggressive drivers had more anxious feelings when interacting with AVs compared to HVs. Driving simulator studies [8, 16, 29, 38, 44] and field experiments [20, 30] were conducted to characterize the behavioral adaptation of human drivers when interacting with AVs. These studies showed that human drivers performed a higher speed and acceleration [20], shorter lane changing duration [16, 29], smaller time headways [8, 29, 30, 38], higher time-to-collision [20], and higher gap acceptance [44] when interacting with AVs compared to HVs.

The HV-AV interaction was explored in various traffic scenarios, such as merging, lane changing, and car-following scenarios. Some studies constructed the one-on-one interaction scenarios between HV and AV [19, 20, 30, 44], while others focused on the interaction between HVs and AV platoons [8, 16, 29, 38]. For example, Rad et al. [29] constructed three different traffic scenarios (base: only HVs; mixed: platoons of 2–3 AVs driving on any lane and mixed with HVs; dedicated lane: platoons of 2–3 AVs driving only on a dedicated lane) in a fixed driving simulator. Their results indicated that in a dedicated lane scenario, drivers in HVs maintained shorter distances behind other cars when following them and were willing to take smaller gaps when changing lanes, as opposed to in mixed and base scenarios. A series of studies by Tanimoto et al. [24, 41, 42] suggested that social dilemmas may underlie a traffic

flow phenomenon. They found that the structures of some traffic cases, including 2 into 1 lane junction (bottleneck), lane changes, and route selection, corresponded to the chicken game and public goods games in game theory. However, little research has employed game theory to theoretically categorize driving scenarios and investigate their effects on drivers' behaviors and decision-making in mixed-traffic environments.

Almost all the studies above are based on the assumption that AVs have defensive programming and are designed to be law-abiding. Perceiving AVs as being defensive, safe, and compliant with traffic laws may encourage people's intention to take advantage of AVs on the road. Autonomous driving companies have reported that individuals, including pedestrians and human drivers, displayed aggressive behavior towards AVs while they were being tested on public roads [9, 32]. This could result in potential interaction conflicts. AVs have to learn to be aggressive in the right amount according to culture [35]. Although studies have shown that AVs' driving styles affected human trust and acceptance when driving or riding in AVs [5, 18], little has been done to investigate how AVs' driving styles influence human drivers' decision-making in HVs when they interact with AVs on the road.

Game theory has revealed numerous aspects of human cooperative behavior and can be used as a framework to study mutual cooperation. Social dilemmas such as prisoners' dilemma [12, 27, 46], chicken game [14, 43], and public goods game [4] have been adopted to explore human-machine cooperation. For example, Paeng et al. [27] and Wu et al. [46] investigated whether the individual's decisions were influenced by the agent type they interacted with in a prisoner's dilemma. Their results showed that humans had a greater degree of trust in robots rather than other humans. Ishowo-Oloko et al. [12] engaged participants in a repeated prisoner's dilemma game and provided them with either accurate or inaccurate information about their partner's nature type. The result indicated that robots outperformed humans in promoting cooperation at inducing cooperation without disclosing their true nature. Similarly, in the study of Kim et al. [14], human players altered their behaviors in response to the agent's behavioral pattern in the iterative chicken game and human players were more sensitive to fairness when they were told to play with a human. While behavioral game theory shows great potential to study human-robot interaction, little has been done to apply game theory to investigate human behavior and decision-making in HV-AV interactions in mixed traffic.

3 METHOD

We developed an online survey to explore how driver driving styles, vehicle driving styles, interaction types, scenario types, and time constraints impacted driver decisions in mixed-traffic environments. This research complied with the American Psychological Association Code of Ethics and was approved by the institutional review board at the University of Pittsburgh.

3.1 Experimental design

Independent variables. We constructed a study with a $2 \times 2 \times 2 \times 2 \times 2$ mixed factorial design. The between-subjects variables were the driving styles of participants and interacting vehicles (aggressive vs. conservative). An aggressive driving style is associated with

frequent changes in throttle and brake pedal positions. Aggressive drivers drive with sharp and abrupt accelerations and decelerations [21]. A conservative driving style is associated with small amplitudes and low-frequency changes in the accelerator and brake pedal. Conservative drivers exhibit mild operational behaviors [21]. The within-subjects variables were interaction types (HV-HV vs. HV-AV), time constraints (high vs. low), and scenario types (chicken game vs. public goods game). In our experiment, the automation level of AVs is SAE Level 5, which means that the automated driving system can drive the vehicle under all conditions and humans are not required to perform all driving tasks [11]. Under high time constraints, drivers were requested to reach a destination with utmost haste, whereas they drove without time pressure under low time constraints. All variables were binary categorical variables. Each condition was designed using a text description and an animated scenario (see Figure 1) to help participants understand the context. The order of each condition was random across participants.



Figure 1: An example of survey questions

Dependent variable. The dependent variable was the participants' decisions. After participants watched each scenario animation, they were given two options (see Table 1 and Figure 1). Choosing the first option indicates a relatively aggressive action, and choosing the second one indicates a relatively conservative action.

Scenarios. We constructed seven game theory scenarios (see Table 1) to test driver decisions in mixed-traffic environments. We created scenarios from a top-down viewpoint to enhance the understanding of the overall traffic situations. The chicken game is an experiment used to analyze conflict situations in which two players can cooperate or compete. Cooperative one is deemed the "chicken" and the other one wins [39]. In the chicken game scenarios, the subject car (the car participants were driving) and another car competed for the right of way, a parking lot, or the priority of merging. Public goods game is a multiplayer experiment used to study the behavior of individuals in a group setting: those who cooperate contribute to the public good, while those who defect do not make any contribution. The total contribution is multiplied by an enhancement factor less than the number of members, and the result is distributed evenly among all members of the group. Hence, defectors get the same benefit as cooperators at no cost [36]. In the public goods game scenarios, the subject car in the traffic had two strategies to compete: changing lanes aggressively to overtake other vehicles or keeping in the current low-speed lane. In both

types of game theoretical scenarios, players had the option to either cooperate (take a conservative action) or defect (take an aggressive action) to achieve their benefits such as efficiency and safety.

3.2 Procedure

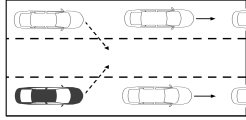
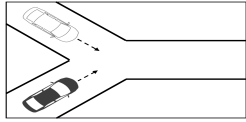
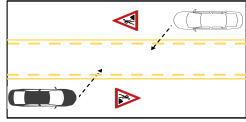
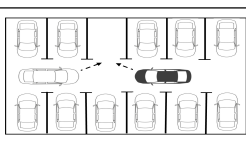
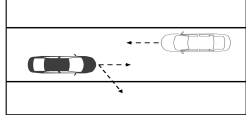
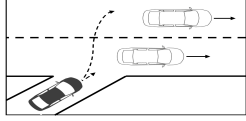
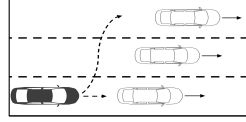
The consent, instructions, driving scenarios, and questionnaires were integrated into Qualtrics, and the study was launched on the CloudResearch platform. First, participants who met the criteria were randomly assigned to interact with vehicles with either aggressive or conservative driving styles. Then they were guided to read the instructions and take the training session. Participants were required to read the introduction of scenarios and answer the attention check questions carefully. If they failed the attention check, the survey would end automatically. After the training session, participants were asked to imagine themselves as the driver of the HV and make decisions when interacting with other vehicles in eight conditions (2 interaction types $\times$ 2 types of time constraints $\times$ 2 scenario types). For each condition, they were asked to read a text description and watch an animated traffic scenario, and then indicate their preferred decision from two choices. Each participant encountered 4 chicken game scenarios and 4 public goods scenarios. These scenarios were created by random sampling with replacement from 5 chicken game scenarios and 2 public goods game scenarios. In the end, participants reported their demographic information (e.g., gender, age, years of driving), and answered the violation subscale of Driving Behavior Questionnaire (DBQ) [33].

3.3 Participants

A total of 312 people were recruited from the CloudResearch platform and they were all from the United States. We obtained informed consent from each participant. We screened them for various inclusion criteria including 18 years or older, driver's license status, and non-colorblind. We deleted the data of participants who timed out (we set the completion time as 80 minutes), submitted in less than 5 minutes using the incorrect link, or failed the attention check. At last, 252 qualified people (126 males, 123 females, 2 transgender women, and 1 prefer not to answer) participated in the study. The participants' ages ranged from 18 to 79 with an average age of 40.1 years ($SD=11.5$ years). Participants were paid \$4 for their participation in the 15-min online survey study.

We characterized drivers' driving styles into two categories based on the scores in the adapted violation subscale of DBQ. Due to social desirability bias, a cautious attitude should be taken towards the self-reports of the alcohol-related item [15]. Given the different effects of alcohol in different regions [25, 40], we removed it and kept the other 11 items. All the items were rated on six-point rating scales (0: never; 5: nearly all the time). Zhao et al. [47] indicated drivers with high DBQ violation scores showed more aggressive objective driving behavior. By looking at the relative interquartile ranges [10], we divided our participants into two groups: relatively aggressive (116 participants; scores: $\mu = 25.96, \sigma = 5.16$) and relatively conservative (136 participants; scores: $\mu = 16.26, \sigma = 2.81$). For the between-subjects variable, vehicle driving styles, 125 participants interacted with vehicles with conservative driving styles and 127 participants interacted with vehicles with aggressive driving styles.

Table 1: Game theoretic driving scenarios

Abstract game	Scenario descriptions	Possible decisions	Diagrams
Chicken game	Two vehicles in the most-right and most-left lanes of a three-lane road, are slowed down by the traffic ahead, and plan to switch to the empty middle lane.	(1) switch to the empty middle lane first, or (2) yield to another vehicle and switch after it at the cost of a time delay.	
	Two vehicles on two equal-priority merging-into-one roads trying to join the single road (like an inverted fork).	(1) merge into one road first, or (2) yield to another vehicle and then merge.	
	On a three-lane, two-way road, two vehicles are running in the most-right and most-left lanes in the opposite directions. They both need to encroach on the same middle lane to pass an obstruction.	(1) try to change to the middle lane first, or (2) wait until the other vehicle passes	
	Two vehicles are competing for a parking lot.	(1) speed up and occupy the parking lot, or (2) give up the parking lot.	
	Two vehicles are heading towards each other on a narrow two-way traffic road	(1) continue forward assuming the opposing vehicle will yield, or (2) yield to the opposing vehicle by reversing or swerving onto the sidewalk	
Public Goods Game	A vehicle is joining a two-lane road with the aim of turning right at the next exit. The right-line is a right-turn-only lane and is the slower lane of the two.	(1) join the (slower) right lane first, or (2) continue ahead on the (faster) left lane and make its way in front of other vehicles on the right-hand lane.	
	Driving in traffic congestion.	(1) adhere to speed limits and remain in your lane, or (2) swerve aggressively between lanes when gaps between vehicles emerge.	

3.4 Data Analysis

Statistical analysis was performed using IBM SPSS Statistics Version 28.0. Due to the repeated measurements and the binary responses, Generalized Estimating Equations (GEE) were constructed to analyze the effects of independent variables (driver and vehicle driving styles, interaction types, time constraints, and scenario types) on the dependent variable (driver decisions). Binomial was specified as distribution and Logit was specified as the link function.

4 RESULTS

The test of model effects showed that there were significant main effects of time constraints ($\chi^2(1) = 123.357; p < .001$) and driver driving styles ($\chi^2(1) = 8.735; p = .003$) on driver decisions. There were significant interaction effects between vehicle driving styles

and scenario types ($\chi^2(1) = 4.423; p = .035$), between time constraints and scenario types ($\chi^2(1) = 6.229; p = .013$), and between interaction types and driver driving styles ($\chi^2(1) = 3.425; p = .064$). All the other effects were not significant. Below we further explain the interaction effects in detail.

The significant interaction effect between vehicle driving styles and scenario types on driver decisions (see Figure 2a) suggested that in the chicken game scenarios, drivers had a higher chance of taking aggressive actions when interacting with vehicles with conservative driving styles than aggressive driving styles ($p = .008$). Yet, in the public goods game scenarios, the effect of driving style on driver decision-making was not significant ($p = .977$).

The significant interaction effect between time constraints and scenario types (see Figure 2b) showed that under low time constraints (not in a hurry to reach a destination), drivers in the chicken

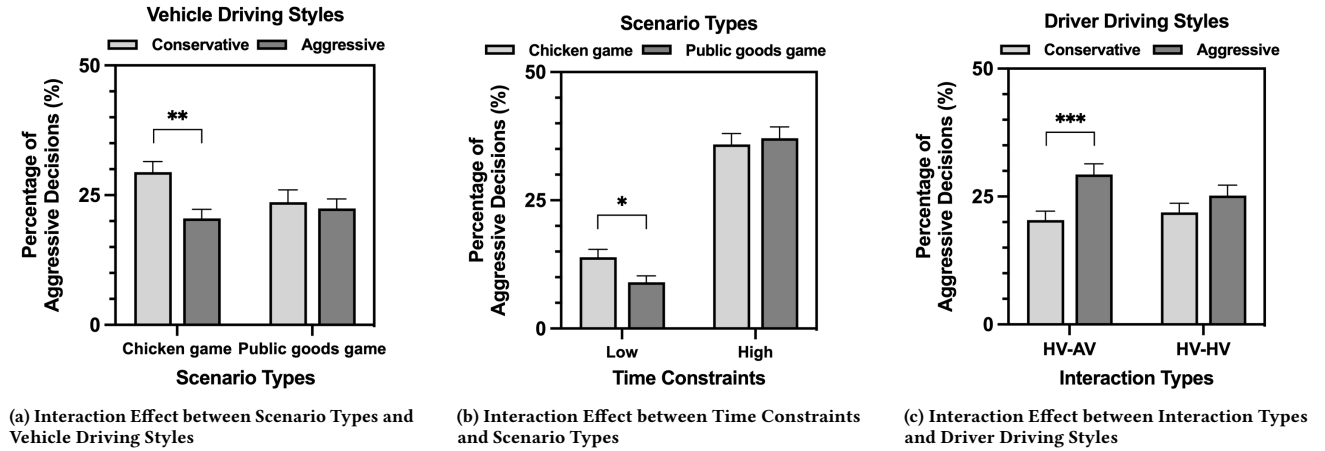


Figure 2: The interaction effect results

game scenarios were more likely to take aggressive actions than drivers in the public goods game scenarios ($p = .016$). In contrast, under high time constraints, scenario types did not have a significant effect on driver decision-making ($p = .587$).

The borderline significant interaction effect between interaction types and driver driving styles (see Figure 2c) indicated that when interacting with AVs, relatively aggressive drivers had a higher likelihood of taking aggressive actions than relatively conservative drivers ($p < .001$). However, when interacting with HVs, the effect of driver driving styles on their decision-making was not significant ($p = .105$).

5 DISCUSSION

This study investigated the impacts of driver and vehicle driving styles (aggressive vs. conservative), interaction types (HV-HV vs. HV-AV), scenario types (chicken game vs. public goods game scenarios), and time constraints (high vs. low) on driver decisions (take an aggressive action vs. take a conservative action) in mixed-traffic environments.

Our results suggested that drivers were more likely to exhibit aggressive behaviors when faced with a high time constraint. Different from prior work by Trendelenburg et al. [44], we found that the effects of time constraints were similar in HV-AV interaction and HV-HV interaction. One possible explanation for this discrepancy could be that their participants were informed that AVs drove more cautiously than HVs but our study treated AVs and HVs the same. Additionally, our work considered more situations and our results were more generalizable. For example, our participants were exposed to a variety of scenarios, while their participants only experienced a specific right and left turn at an intersection scenario. Also, our participants were from different age and occupation groups but their participants were relatively young with an academic background.

Our results indicated that aggressive drivers were more likely to take aggressive actions than conservative drivers and driving styles had a greater impact on HV-AV interaction than HV-HV interaction. Specifically, aggressive drivers took more aggressive decisions compared to conservative drivers when interacting with AVs but such

differences did not exist in HV-HV interaction. This may be because aggressive drivers and conservative drivers perceived the risk of AVs differently. We did not find a significant difference of decisions towards AVs and HVs among aggressive drivers or among conservative drivers, which seemed different from Ma and Zhang [19]. Their results suggested that aggressive drivers were more likely to take advantage of interacting vehicles in HV-AV interaction than in HV-HV interaction. One big difference between these two studies is that the driving styles of AVs could be either aggressive or conservative in our study but were coded to drive cautiously in their study. Our results clarified the previous work and indicated that drivers may take advantage of interacting vehicles with conservative driving styles rather than take advantage of AVs.

To expand the existing studies, we used the chicken game and public goods game in game theory to develop two types of driving scenarios. Our results showed that in the chicken game scenarios, drivers took a higher proportion of aggressive actions when interacting with vehicles with conservative driving styles than aggressive driving styles. However, in public goods game scenarios, the driving styles of other vehicles did not have a significant impact on driver decision-making. This may be because in one-to-one interactions (chicken game scenarios), drivers perceived their own aggressive behavior as less risky when interacting with vehicles with conservative driving styles. Meanwhile, we found that drivers were more likely to take conservative actions in the public goods game scenarios than in chicken game scenarios under low time constraints, whereas there was no such difference under high time constraints. In our settings, high time constraints (reaching to the destination as quickly as possible) represented temporal advantages, whereas low time constraints meant no apparent benefits. Our results indicated that drivers might be more prone to taking actions for collective benefit and had more intention to cooperate in the traffic flow when there was no incentive in one-to-many interactions. Our findings were consistent with previous literature, which revealed that individuals frequently cooperated with unrelated strangers in large groups when there were minimal or no potential gains in terms of reputation [7].

This study highlights the need to design AVs with different driving styles and optimize their performance and strategies according to human decision-making patterns in traffic scenarios. Our results have implications to develop driver behavior prediction systems, which could predict driver behaviors and decision-making when interacting with AVs in novel situations. It also indicates the importance of developing innovative communication methods between HVs and AVs. For example, human-machine interfaces could be designed to inform drivers of the intentions of AVs and provide recommendations on driving strategies.

6 LIMITATIONS & FUTURE WORK

There are some limitations that we could address in the future. First, our sample was geographically limited to the United States, which may not represent the perceptions and behaviors of other populations in distinct regions. To gain a more comprehensive understanding, cross-country research would be necessary in the future. For example, future work could collect data from China, which is very culturally different from the US, based on WEIRDness distance [23]. The use of different countries will help establish the external validity of the findings across a wide-set of countries. Second, our game theoretical scenarios were described using text and animations, and participants' decisions were collected from web-based questionnaires. In such settings, we were only able to investigate participants' hypothetical responses. Open-ended questions could be incorporated to elicit feedback from participants regarding their identification of distinct scenarios as manipulation checks. Future research can examine participants' revealed decisions through simulator studies and on-road testing. More dependent variables, such as drivers' driving strategies and performance in the dynamic environment could be collected to indicate drivers' cooperative behaviors. Third, future research could develop more scenarios and manipulate more interaction settings (e.g., number of passengers, vehicle automation level, vehicle brand) and driving environments (e.g., geographical locations, weather, and light conditions) to investigate human-machine cooperation in mixed-traffic environments. This will be crucial to (1) establish replicability of the results, (2) study cultural generalizability, and (3) heterogeneous effect analysis.

7 CONCLUSION

This study investigated how the driver and vehicle driving styles, interaction types, scenario types, and time constraints influenced human drivers' decision-making in mixed-traffic environments. First, we found that in general drivers with aggressive driving styles and time constraints were more likely to take aggressive actions. Second, in the chicken game scenarios, drivers had a higher chance of taking aggressive actions when interacting with vehicles with conservative driving styles than aggressive driving styles. Yet, such differences did not exist in the public goods game scenarios. Third, drivers were more likely to take aggressive actions in the chicken game scenarios than in the public goods game scenarios under low time constraints, whereas there was no such difference under high time constraints. Fourth, relatively aggressive drivers had a higher likelihood of taking aggressive actions than relatively conservative drivers when interacting with AVs but such differences did not exist in HV-HV interaction. Our findings provide essential insights into

the design of AVs and promote the development of related laws and policies to facilitate human-machine cooperation in mixed-traffic environments.

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